Answers and talk notes

Determine the Whole Given the Part and Percent

What your student is learning

Tonight we write a percent equation — part = percent × whole — and solve it for the unknown whole by dividing.

Answers

Ask your student to explain before you look.

1. **C.** 0.4 × n = 12. Percent times whole equals part, so 0.4 × n = 12.
2. **A.** Divide 8 by 0.5. Divide 8 by 0.5. Division undoes multiplication, so w = 8 ÷ 0.5.
3. **D.** 14. w = 7 ÷ 0.5 = 14, because there are 2 halves in each whole.
4. **B.** 90. w = 9 ÷ 0.1 = 90, because there are 10 tenths in each whole.
5. **C.** 24. w = 6 ÷ 0.25 = 24, because there are 4 fourths in each whole.
6. **A.** $45. 0.2 × b = 9, so b = 9 ÷ 0.2 = 45. The bill was $45.

Talk about it

Is the whole bigger or smaller than the part, and how does your check show the answer is right?

Home mission

Use a real receipt. Pretend the tip was $5 and it was 20% of the bill. Can your student write 0.2 × b = 5 and solve it? Then check: is 20% of the answer really $5?

When they are stuck, ask

- Which part of "percent × whole = part" is missing here?
- Should the whole be bigger or smaller than the part?
- Can you check by finding that percent of your answer?

Try not to say

- Multiply the part by the percent to find the whole.
- Just move the decimal point and you are done.
- Checking is optional if the answer looks fine.